SQA Automation - Test Case Document

Project: Demo Web Shop (tricentis.com) | **Environment:** Windows 11, Chrome/Edge
Framework: TestNG + Selenium WebDriver + Page Object Model | **Report Date:** Jun 29, 2025

Test Summary

Metric	Value
Total Test Cases	15
Passed	15 (100%)
Failed	0 (0%)
Total Duration	2 min 9 sec

Total: 15 | **Passed: 15** | **Failed: 0**

1. Registration Tests

TC-01: validRegistrationTest

Field	Details
Test ID	TC-01
Module	Registration
Test Name	Verify successful user registration with valid data
Preconditions	User is on the home page
Test Data	firstName = "TestFirstName", lastName = "User", email = RandomEmail, password = "testpass123"
Steps	1. Click Register link on HomePage 2. Fill registration form: gender (Male), first name, last name, email, password, confirm password 3. Click Register button 4. Verify success message is displayed
Expected Result	Registration success message ("Result") is displayed
Actual Result	Pass - Success message displayed
Status	PASS

TC-02: existingEmailRegistrationTest

Field	Details
Test ID	TC-02
Module	Registration
Test Name	Verify registration with an already-existing email shows error
Preconditions	User is on the home page
Test Data	firstName = "Test", lastName = "User", email = RandomEmail (reused), password = "testpass123"
Steps	1. Click Register link on HomePage 2. Register a new user with a randomly generated email 3. Click Continue after successful registration 4. Click Log out to return to the home link 5. Click Register link again 6. Register with the SAME email as step 2 7. Verify error message is displayed
Expected Result	Error message ("_message-error") should be displayed indicating duplicate email
Actual Result	Pass - Error message displayed for duplicate email
Status	PASS

2. Login Tests

TC-03: validLoginTest

Field	Details
Test ID	TC-03
Module	Login
Test Name	Verify successful login with valid credentials
Preconditions	Registered user exists (created within test)
Test Data	email = RandomEmail, password = "testpass123"
Steps	1. Register a new user (firstName="Test", lastName="User", random email) 2. Log out via logout link 3. Click Log in link 4. Enter email and password, click Log in 5. Verify user is logged in and email matches
Expected Result	User is logged in (account link displayed) and email text matches registered email
Actual Result	Pass - Login successful, email matches
Status	PASS

TC-04: invalidLoginTest

Field	Details
Test ID	TC-04
Module	Login
Test Name	Verify error message for invalid login credentials
Preconditions	User is on the home page
Test Data	email = "invalid@test.com", password = "wrongpass"
Steps	1. Click Log in link on HomePage 2. Enter invalid email and invalid password 3. Click Log in button 4. Verify login error message is displayed
Expected Result	Error message ("_message-error") should be displayed for invalid credentials
Actual Result	Pass - Error message displayed
Status	PASS

TC-05: emptyCredentialsTest

Field	Details
Test ID	TC-05
Module	Login
Test Name	Verify login fails with empty credentials and stays on login page
Preconditions	User is on the home page
Test Data	email = "", password = ""
Steps	1. Click Log in link on HomePage 2. Leave email and password fields empty 3. Click Log in button 4. Verify URL still contains "login" (not redirected away)
Expected Result	Current URL should contain "login" (user is not redirected away)
Actual Result	Pass - URL contains "login"
Status	PASS

3. Product Tests

TC-06: searchExistingProductTest

Field	Details
Test ID	TC-06
Module	Product Search
Test Name	Verify search returns results for an existing product
Preconditions	User is on the home page
Test Data	searchQuery = "computer"
Steps	1. Type "computer" in the search box 2. Click Search button 3. Verify product list is displayed
Expected Result	At least one product item (".product-item") should be displayed
Actual Result	Pass - Product list displayed
Status	PASS

TC-07: searchInvalidProductTest

Field	Details
Test ID	TC-07
Module	Product Search
Test Name	Verify search returns no results for a non-existent product
Preconditions	User is on the home page
Test Data	searchQuery = "xyz123invalid"
Steps	1. Type "xyz123invalid" in the search box 2. Click Search button 3. Verify NO products are displayed
Expected Result	No product items should be displayed (list is empty)
Actual Result	Pass - No products displayed
Status	PASS

TC-08: viewProductDetailsTest

Field	Details
Test ID	TC-08
Module	Product Browsing
Test Name	Verify clicking a product navigates to the product details page
Preconditions	Navigate to Books page
Test Data	productIndex = 0 (first product)
Steps	1. Navigate to Books category 2. Click on the first product 3. Verify product detail page is displayed
Expected Result	Product detail page (".product-details-page") element should exist
Actual Result	Pass - Product detail page displayed
Status	PASS

4. Cart Tests

TC-09: addProductToCartTest

Field	Details
Test ID	TC-09
Module	Shopping Cart
Test Name	Verify a product can be added to the cart
Preconditions	Navigate to Books page
Test Data	productIndex = 0 (first product)
Steps	1. Navigate to Books category 2. Open first product details 3. Click Add to cart button 4. Wait for notification bar 5. Navigate to cart page 6. Verify cart has at least 1 item
Expected Result	Cart item count should be greater than 0
Actual Result	Pass - Product added to cart
Status	PASS

TC-10: removeProductTest

Field	Details
Test ID	TC-10
Module	Shopping Cart
Test Name	Verify a product can be removed from the cart
Preconditions	Product already added to cart
Test Data	productIndex = 0 (first product)
Steps	1. Navigate to Books, add first product to cart 2. Navigate to cart page 3. Record current item count 4. Check the "Remove" checkbox for the product 5. Click "Update shopping cart" button 6. Verify cart is empty or has fewer items
Expected Result	Cart should be empty OR item count should be less than before removal
Actual Result	Pass - Product removed from cart
Status	PASS

TC-11: updateQuantityTest

Field	Details
Test ID	TC-11
Module	Shopping Cart
Test Name	Verify product quantity can be updated in the cart
Preconditions	Product already added to cart
Test Data	productIndex = 0, newQuantity = "3"
Steps	1. Navigate to Books, add first product to cart 2. Navigate to cart 3. Clear quantity field and type "3" 4. Click "Update shopping cart" button 5. Verify page source contains "3"
Expected Result	Page source should contain the updated quantity "3"
Actual Result	Pass - Quantity updated to 3
Status	PASS

5. Checkout Tests

TC-12: checkoutEmptyCartTest

Field	Details
Test ID	TC-12
Module	Checkout
Test Name	Verify empty cart message is displayed when accessing checkout with no items
Preconditions	Navigate to /cart (no items added)
Test Data	None
Steps	1. Navigate directly to /cart page 2. Verify empty cart message is displayed
Expected Result	Cart should be empty (".order-summary-content" contains "Your Shopping Cart is empty!")
Actual Result	Pass - Empty cart message displayed
Status	PASS

TC-13: successfulCheckoutTest

Field	Details
Test ID	TC-13
Module	Checkout
Test Name	Verify complete end-to-end checkout flow with valid details
Preconditions	User not logged in
Test Data	**Registration:** firstName = RandomName, lastName = "User", email = RandomEmail, password = "testpass123" **Billing:** country = "Sri Lanka", state = "Eastern", city = "Batticaloa", address = "123 Main St", zip = "30000", phone = "0345678900" **Shipping:** Ground (index 0) **Payment:** Cash on Delivery (index 0)
Steps	1. Register a new user 2. Navigate to Books, add first product to cart 3. Navigate to cart, click "I agree" and Checkout 4. Fill billing address with all required fields 5. Click Continue (billing) - handle overlapping address step 6. Select Ground shipping method, click Continue 7. Select payment method (Cash on Delivery) and click Continue 8. Click Continue on payment info 9. Click Confirm 10. Verify order completed message
Expected Result	Order completion message ("Your order has been successfully processed") should be displayed
Actual Result	Pass - Order completed successfully
Status	PASS

6. Contact Tests

TC-14: submitContactFormTest

Field	Details
Test ID	TC-14
Module	Contact Us
Test Name	Verify contact us form can be submitted successfully
Preconditions	Navigate to /contactus page
Test Data	name = RandomName, email = RandomEmail, enquiry = "This is a test enquiry."
Steps	1. Navigate to Contact Us page 2. Fill in name, email, and enquiry fields 3. Click Submit button 4. Verify success message is displayed
Expected Result	Success message (".result" - "Your enquiry has been sent") should be displayed
Actual Result	Pass - Enquiry sent successfully
Status	PASS

7. Account Tests

TC-15: myAccountPageTest

Field	Details
Test ID	TC-15
Module	Account
Test Name	Verify account page shows correct logged-in email after registration
Preconditions	User is on the home page
Test Data	firstName = "Test", lastName = "User", email = RandomEmail, password = "testpass123"
Steps	1. Click Register link 2. Register with valid details 3. Click Continue after registration 4. Click My account link 5. Assert the displayed email matches the registered email
Expected Result	User should be logged in (account link displayed) and the account email should match the registered email
Actual Result	Pass - Account page shows correct email
Status	PASS

Environment & Configuration

Parameter	Value
Operating System	Windows 11
Browser(s)	Chrome / Edge
Target Application	https://demowebshop.tricentis.com/
Automation Framework	Selenium WebDriver + TestNG
Design Pattern	Page Object Model (Page Factory)
Reporting	ExtentReports (Spark)
Data Generation	RandomDataUtility (random names/emails)